**1** Expands brackets correctly: $12x - 4 - 5x \geq 2x + 16$ **[M1]**

Rearranges to collect $x$ terms: $7x - 2x \geq 16 + 4$ or $5x \geq 20$ **[M1]**

$x \geq 4$ **[A1]**

*[Total Marks: 3]*

**2 (a)** $3x - 12 < 5x - 2$

**[M1]**

$-10 < 2x \implies x > -5$ (or $-5 < x$)

**[A1]**

**(b)** $-4$

**[B1]**

*[Total Marks: 3]*

**3 (a)** $-6 < 3x \leq 12$ **[M1]**

$-2 < x \leq 4$ **[A1]**

**(b)** $-1, 0, 1, 2, 3, 4$ **[B2]**

(or [B1] for 5 correct with no incorrect, or all 6 correct with one incorrect)

*[Total Marks: 4]*

**4 (a)** $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

**(b)** Open circle at $-2$ and closed circle at 4 **[B1]**

Thick line joining the two circles **[B1]**

*[Total Marks: 4]*

**5 (a)** $2x < 12$ **[M1]**

$x < 6$ **[A1]**

**(b)** Recognises open circle means strict inequality ($>$ or $<$) and arrow to right means greater than **[M1]**

$x > -2$ **[A1]**

*[Total Marks: 4]*

**6 (a)** Expands brackets correctly: $3x - 6 \leq 5x + 8$ **[M1]**

Rearranges to isolate $x$, e.g. $-14 \leq 2x$ or $-2x \leq 14$ **[M1]**

$x \geq -7$ **[A1]**

**(b)** Places a closed, solid circle at $-7$ **[B1]**

Draws a line with an arrow pointing to the right from the circle **[B1]**

*[Total Marks: 5]*

**7 (a)** $-9 > 2x$ or $-2x > 9$ **[M1]**

$x < -4.5$ **[A1]**

**(b)** $-4 < 2n \leq 4$ **[M1]**

$-2 < n \leq 2$ **[M1]**

$-1, 0, 1, 2$ **[A1]**

**(c)** $x > -1$ (accept any valid variable) **[B1]**

*[Total Marks: 6]*

**8 (a)** $-15 \leq 3x$ or $-3x \leq 15$ **[M1]**

$x \geq -5$ or $-5 \leq x$ **[A1]**

**(b)** $-7 < 2x \leq 8$ **[M1]**

$-3.5 < x \leq 4$ **[A1]**

**(c)** Largest $= 4$ **[B1]**

Smallest $= -3$ **[B1]**

*[Total Marks: 6]*

**9 (a)** $-3x < 9$ or $2 - 11 < 3x$ **[M1]**

$x > -3$ **[A1]**

**(b)** $8x - 4 \leq 5x + 17$ **[M1]**

$3x \leq 21$ **[M1]**

$x \leq 7$ **[A1]**

**(c)** $-3 < n \leq 7$ **[M1]**

$-2, -1, 0, 1, 2, 3, 4, 5, 6, 7$ **[A1]**

*[Total Marks: 7]*

**10 (a)** Subtracts $3x$ and 3 to reach $\pm 4x \leq \pm 8$ or similar **[M1]**

$x \geq -2$ **[A1]**

**(b)** Solid circle (or equivalent clear inclusive mark) placed at $-2$ **[B1]**

Thick line or arrow pointing to the right starting from $-2$ **[B1]**

**(c)** Adds 5 to all parts to get $3 < 3x \leq 9$ **[M1]**

Divides by 3 to get $1 < x \leq 3$ **[A1]**

$2, 3$ **[A1]**

*[Total Marks: 7]*

**11** $y \leq 5x - 4$ or equivalent **[B1]**

$2y \geq x + 1$ or equivalent **[B1]**

$x + y \leq 8$ or equivalent **[B1]**

*[Total Marks: 3]*

**12** $x \geq 2$ **[B1]**

$y > x$ **[B1]**

$x + y \leq 8$ (or $y \leq -x + 8$) **[B1]**

*[Total Marks: 3]*

**13** $x \geq 1$

**[B1]**

$y \geq 2$

**[B1]**

$x + y < 6$ (or $y < -x + 6$)

**[B1]**

*[Total Marks: 3]*

**14 (a)** $x < 4$ **[B1]**

$y \geq -x + 3$ **[B1]**

$y \leq \frac{1}{2}x + 3$ (or equivalent, e.g. $2y \leq x + 6$) **[B1]**

**(b)** Substitutes $x = 3$ into the inequalities: $b \geq -3 + 3 = 0$ and $b \leq 0.5(3) + 3 = 4.5$ **[M1]**

Deduces the range $0 \leq b \leq 4.5$ **[A1]**

$0, 1, 2, 3, 4$ **[A1]**

*[Total Marks: 6]*

**15** Identifies horizontal line $y = 2$ **[M1]**

States $y \geq 2$ (accept $y > 2$) **[A1]**

Identifies dashed line passing through $(0, 6)$ and $(4, 2)$ **[M1]**

Finds equation $y = -x + 6$ and states $y < -x + 6$ **[A1]**

Identifies solid line passing through $(0, 2)$ and $(1, 5)$ **[M1]**

Finds equation $y = 3x + 2$ and states $y \leq 3x + 2$ **[A1]**

*[Total Marks: 6]*

**16** **(a)** Multiplies by 15 to clear denominators: $3(3x-2)-5(2x-5) \leq 30$ **[M1]**

$9x-6-10x+25 \leq 30$ **[M1]**

$-x+19 \leq 30$ **[M1]**

$x \geq -11$ **[A1]**

**(b)** Identifies that $y$ can only take integer values of 2 or 3 (since $2y > x+y > 2 \implies y > 1$) **[M1]**

Finds correct points for one of the valid $y$ values: e.g. $(1,2)$ for $y=2$ OR $(0,3),(1,3),(2,3)$ for $y=3$ **[B1]**

States all four correct points: $(1,2),(0,3),(1,3),(2,3)$ with no extras **[A1]**

*[Total Marks: 7]*

**17** $y \geq 2$ **[B1]**

$y \leq 2x$ **[B1]**

$x+y \leq 6$ **[B1]**

*[Total Marks: 3]*

**18** $x \geq 1$ **[B1]**

$y \leq 2x$ **[B1]**

$x+y \leq 6$ **[B1]**

(Note: Accept strict inequalities $<,>$)

*[Total Marks: 3]*

**19** **(a)** Adding 1 to all parts: $-6 \leq 3x < 12$ **[M2, M1 for one side correct]**

Dividing by 3: $-2 \leq x < 4$ **[A1]**

**(b)** Identifies at least 3 correct integers or includes 4 but excludes -2 **[M1]**

$-2,-1,0,1,2,3$ **[A1]**

*[Total Marks: 5]*

**20** **(a)** $y \geq 1$ **[B1]**

$y \leq x+1$ **[B1]**

$y < 5-x$ (or $x+y<5$) **[B1]**

All three inequality signs correct (with strict inequality for the dashed line) **[B1]**

**(b)** Identifies $(1,2)$ as a valid integer coordinate point inside $R$ **[M1]**

$(1,2)$ **[A1]**

*[Total Marks: 6]*

**21** **(a)** Subtracts 1 from all parts: $-6 < 2x \leq 10$ **[M1]**

Divides by 2 across all parts **[M1]**

$-3 < x \leq 5$ **[A1]**

**(b)** Identifies at least two correct integers from the set **[M1]**

$-2, -1, 0$ **[A1]**

**(c)** Finds the correct roots of the quadratic $2x^2 - 7x + 3 = 0$ as 0.5 and 3 **[B1]**

$x < 0.5$ or $x > 3$ **[B1]**

*[Total Marks: 7]*

**22** **(a)** $y \geq 1$ **[B1]**

$y \leq 2x - 2$ **[B2]**

(or $M1$ for finding the boundary equation $y = 2x - 2$)

$x + y \leq 7$ (or $y \leq -x + 7$) **[B2]**

(or $M1$ for finding the boundary equation $x + y = 7$)

**(b)** Identifies a valid integer coordinate inside or on the boundary of $R$ that maximizes the expression, e.g., $(6, 1)$, $(5, 2)$, or $(4, 3)$ **[M1]**

13 (from $2(6) + 1 = 13$) **[A1]**

*[Total Marks: 7]*

**23** **(a)** $x \geq 15$ **[B1]**

$x + y \leq 40$ **[B1]**

$y \geq \frac{1}{2}x$ or $2y \geq x$ **[B1]**

**(b)** Identifies the objective function for profit: $P = 400x + 150y$ **[M1]**

Tests the valid integer vertex $(26, 14)$ or evaluates the profit equation showing profit is maximised at the highest possible integer value for $x$ **[M1]**

12500 **[A1]**

*[Total Marks: 6]*

**24** **(a)** Draw correct solid line $y = 2$

**[B1]**

Draw correct solid line $y = x + 2$

**[B1]**

Draw correct solid line $2x + y = 8$

**[B1]**

Correct shading of unwanted regions, leaving the triangle with vertices $(0, 2)$, $(3, 2)$, and $(2, 4)$ unshaded and clearly representing $R$.

**[B1]**

**(b)** Identify integer coordinates $(2, 4)$ or $(3, 2)$ as yielding the maximum value

**[M1]**

14

**[A1]**

*[Total Marks: 6]*

**25 (a)** $y \geq 2$ **[B1]**

$y \leq x$ **[B1]**

$x + y \leq 8$ **[B1]**

**(b)** Identifies at least two valid integer coordinate pairs within the region $R$ (e.g., $(6, 2)$, $(5, 3)$, $(4, 4)$) **[M1]**

Evaluates $2x + 3y$ for at least one critical integer point, e.g., $2(4) + 3(4) = 20$ or $2(5) + 3(3) = 19$ **[M1]**

20 **[A1]**

**(c)** Identifies the correct total number of integer points in $R$ is 9, or identifies the 3 correct points where $x = y$: $(2, 2), (3, 3), (4, 4)$ **[M1]**

$\frac{3}{9}$ or $\frac{1}{3}$ **[A1]**

*[Total Marks: 8]*

**26 (a)** $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

**(b)** Multiplies throughout by 12 (or uses common denominator): $4(2x - 5) - 3(x - 3) > 12$ **[M1]**

Expands and simplifies: $8x - 20 - 3x + 9 > 12 \implies 5x - 11 > 12 \implies 5x > 23$ **[M1]**

5 **[A1]**

**(c)** Constructs correct inequality: $3.20 + 1.50k > 5.00 + 1.10k$ **[M1]**

Solves for $k$: $0.40k > 1.80 \implies k > 4.5$ **[M1]**

5 **[A1]**

*[Total Marks: 8]*

**27 (a)** $x + y \leq 500$ **[B1]**

$y \geq 150$ **[B1]**

$5x + 10y \geq 3000$ (or simplified to $x + 2y \geq 600$) **[B1]**

**(b)** Line drawn correctly for $x + y = 500$ (passing through $(0, 500)$ and $(500, 0)$) **[B1]**

Line drawn correctly for $y = 150$ (horizontal line) **[B1]**

Line drawn correctly for $x + 2y = 600$ (passing through $(0, 300)$ and $(600, 0)$) **[B1]**

Correct region identified and all unwanted regions properly shaded (leaving quadrilateral with vertices $(0, 300), (0, 500), (350, 150), (300, 150)$ unshaded) **[B1]**

**(c)** Identifies the vertex $(350, 150)$ as giving the maximum $x$ value **[M1]**

Children's tickets = 350, Adult tickets = 150 **[A1]**

*[Total Marks: 9]*

**28** **(a)** $x + y \leq 60$ **[B1]**

$x \geq 10$ **[B1]**

$y \geq 0.5x$ (or $2y \geq x$ or $y \geq \frac{1}{2}x$) **[B1]**

**(b)** Draws the line $x + y = 60$ correctly (from $(0, 60)$ to $(60, 0)$) **[B1]**

Draws the line $x = 10$ correctly (vertical line at 10) **[B1]**

Draws the line $y = 0.5x$ correctly (from $(0, 0)$ passing through $(20, 10)$, $(40, 20)$, etc.) **[B1]**

Shades the unwanted regions correctly, leaving the correct triangular region $R$ unshaded **[B1]**

**(c)** Identifies $(40, 20)$ as the vertex for maximum profit (or tests multiple vertices) **[M1]**

2100 **[A1]**

*[Total Marks: 9]*

**29** **(a)** Forms left inequality: $3(2x - 3) < 4(x + 5)$ or equivalent **[M1]**

Simplifies to $6x - 9 < 4x + 20$

x ¡ $\frac{29}{2}$ (or 14.5) [A1]

Forms right inequality: $2(x + 5) \leq 3(3x - 1)$ or equivalent **[M1]**

Simplifies to $2x + 10 \leq 9x - 3$

x $\geq \frac{13}{7}$ [A1]

$\frac{13}{7} \leq x < \frac{29}{2}$ (accept $1.86 \leq x < 14.5$ or equivalent double inequality) **[A1]**

**(b)** **(i)** $25 < 12 + 0.07x$ (or $12 + 0.07x > 25$) **[B1]**

**(ii)** $13 < 0.07x$ **[M1]**

$x > 185.71\ldots$ **[A1]**

186 **[A1]**

*[Total Marks: 9]*

**30** **(a)** $x + y \leq 60$ **[B1]**

$y \geq 10$ **[B1]**

$x \geq 15$ **[B1]**

$y \leq 2x$ **[B1]**

**(b)** Line $x + y = 60$ drawn correctly **[B1]**

Line $y = 10$ drawn correctly **[B1]**

Line $x = 15$ drawn correctly **[B1]**

Line $y = 2x$ drawn correctly **[B1]**

Correct region $R$ indicated with all four unwanted regions shaded **[B1]**

**(c)** Identifies $(20, 40)$ as the vertex for maximum profit **[M1]**

Mountain bicycles = 20 and Road bicycles = 40 **[A1]**

Maximum profit = 2600 **[A1]**

*[Total Marks: 12]*

**31** **(a)** $6x - 3 > 4x + 7$ **[M1]**

$2x > 10$ leading to $x > 5$ **[A1]**

**(b)** Open circle placed at 5 with an arrow pointing to the right **[B1]**

**(c)** 6 **[B1]**

*[Total Marks: 4]*

**32** **(a)** $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

**(b)** $-1, 0, 1, 2, 3, 4$ **[B2]**

1 omission or 1 extra integer **[B1]**

**(c)** **(i)** $2(3x + 2) + 2x < 40$ or $(3x + 2) + x < 20$ **[M1]**

$8x + 4 < 40$

8x ¡ 36

x ¡ $4.5 with outerrors [A1]$

**(ii)** Identifies maximum integer $x = 4$ **[M1]**

Length = 14, Width = 4, Area = 56 **[A1]**

*[Total Marks: 8]*

**33** **(a)** $y \geq 1$ **[B1]**

$y \leq x$ **[B1]**

$x + y \leq 6$ or $y \leq -x + 6$ **[B1]**

**(b)** Tests a valid integer coordinate close to the boundary, e.g., $(5, 1)$, $(4, 2)$, or $(3, 3)$ **[M1]**

17 (from $x = 5, y = 1$) **[A1]**

*[Total Marks: 5]*